

Hedging Bitcoin with Its Own Fear Gauge

Evidence from the BVIV Perpetual on Volatility Forecasts, Funding Costs and 15-Minute VWAP Timing

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Abstract

Bitcoin implied volatility is tradable: a perpetual future on the Volmex bviv index has traded on Bitfinex since April 2024. We use 15-minute data from 1 January 2023 to 3 October 2026: Binance BTC prices, the official Volmex index, and the full Bitfinex record of marks, premia and 2,741 funding settlements. With them we ask how BTC volatility should be forecast when a 30-day implied index is observable, and whether a long-bviv position, possibly timed by a volume-weighted average price (VWAP) signal on 15-minute bars, hedges BTC. HAR augmented with implied variance is the best forecast at one day, significantly better than HAR, and at seven days; the bias-corrected index is best at 30 days. Out of sample, with rules frozen on simulated training data, a minimum-variance bviv hedge reduced the expected shortfall (ES) of a long-BTC book by 19.0% (95% block-bootstrap CI 6.9–33.3%), almost entirely through a handful of crash days. The costs are real. Funding settled at its $\pm 0.25\%$ cap in 41% of 8-hour periods and cost a permanently long contract 24 vol points a year, the perpetual traded a median \$71 a day, and executing one hour late cut the ES reduction to 15.8%. A VWAP ratchet adds about two points of ES reduction, not significantly, and placebos with the same triggers at shifted dates do as well: the gain is hedge size, not timing. A calibrated simulation with an arbitrage-free perpetual and 200 test paths reproduces the forecast ranking and the timing result. bviv perpetuals work as a crash hedge, but funding and capacity, not signal design, decide what that hedge is worth.

1 Introduction

A holder of bitcoin, in spot or through a perpetual swap, faces two kinds of risk: an ordinary random walk, and sudden crashes in which liquidity disappears and implied volatility spikes. Futures hedges remove the first and, with it, the position itself; crash protection has required options. The Volmex bviv index measures 30-day BTC implied volatility in a model-free way (Volmex Labs, n.d.), and it has traded as a USDT-margined perpetual on Bitfinex since April 2024 (CoinDesk, 2024), on gTrade since June 2025 (CoinDesk, 2025a) and on Hyperliquid since September 2026 (CoinDesk, 2026). CME announced Bitcoin Volatility futures in May 2026 (CME Group, 2026).

Whether long volatility hedges long BTC depends on the spot-vol correlation, which in BTC is weak and unstable. Positive return shocks have historically raised volatility more than negative ones (Bouri et al., 2017; Baur and Dimpfl, 2018), an asymmetry that has faded (Takaishi, 2021); the BTC–DVOL correlation flipped positive in early 2023 (CoinDesk, 2023), while bviv trended down through the 2024–25 rally (CoinDesk, 2025b). Equity investors learned that long-volatility hedges look good *ex ante* but are eroded by carry and costs (Whaley, 2013; Alexander et al., 2016; Israelov, 2019), except in genuine crises (Szado, 2009). This raises two questions: what does such a hedge cost on a real crypto venue, and can it be *timed* by a signal every intraday trader watches, price relative to its VWAP?

Contributions. (i) We provide, to our knowledge, the first evidence on hedging BTC with a tradable implied-volatility perpetual, including a reconstruction of its funding rule and of its realised costs. (ii) We compare seven volatility forecasts on the official bviv index. (iii) We test VWAP timing with rules frozen on simulated data, single-path block-bootstrap inference and a *timing placebo* that separates timing information from hedge size. (iv) A calibrated simulation with an arbitrage-free perpetual serves as a controlled experiment. A pipeline regenerates every number from public data.

2 The BVIV index and its perpetual

Index. bviv is a 30-day constant-maturity implied-volatility index that Volmex describes as model-free (Volmex Labs, n.d.); secondary sources list Deribit and OKX options as inputs, and Volmex serves it at

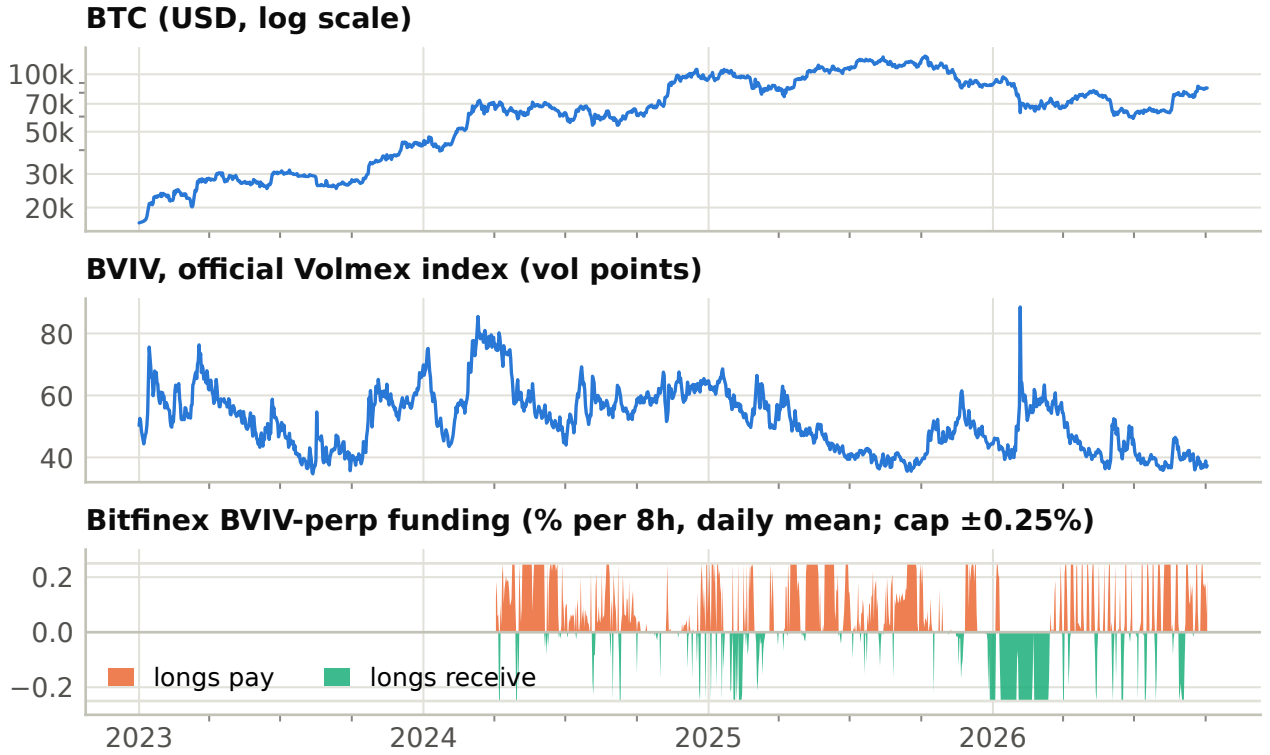


Figure 1: Live data, 1 January 2023 to 3 October 2026. Top: BTC (Binance). Middle: the official Volmex bviv index. Bottom: Bitfinex bviv-perpetual funding per 8-hour period (daily mean), paid by longs when positive.

15-minute resolution. Deribit’s DVOL applies a VIX-style variance-swap calculation (Demeterfi et al., 1999) to Deribit options alone (Deribit, 2021). Implied variance exceeds subsequent realised variance on average: Almeida et al. (2024) estimate about 0.14 a year in variance units for 2017–22, consistent with Alexander and Imeraj (2021) and the equity evidence of Carr and Wu (2009).

The Bitfinex perpetual. One contract pays one dollar (USDt) per index point; the mark price is the index converted at USD/USDt, and over our sample it deviates from the official Volmex index by less than 0.14% on 95% of 15-minute bars. Funding settles every eight hours. The status feed exposes the accrued premium of the perpetual over its mark and two clamp parameters, 0.05% and 0.25%, and the settled rates follow $\text{sign}(a) \min\{\max(|a| - 0.05\%, 0), 0.25\%\}$ of the accrued premium a (Section 3). The cap, 0.75% of notional a day, binds often: in 41% of the 2,741 settlements from April 2024 to October 2026, and long holders paid on 52%. A permanently long contract paid 24 vol points a year on average (51% of notional), but the burden swung from 46 points in 2024 and 34 in 2025 to -12 in 2026, when longs *received* funding (Figure 1). The market is tiny. The perpetual traded on 576 of 914 days, with a median daily volume of \$71 (mean \$7,300) and a median open interest of 84 contracts; its order-book mid sat a median 0.05% above the mark.

3 Data and volatility forecasts

Data. BTC is the Binance BTCUSDt spot market on 15-minute bars (1372 days); quote volume over base volume gives each bar’s exact VWAP. bviv is the official Volmex series from its public 15-minute history. The perpetual’s marks, order-book mids, open interest and accrued funding come from about 1.3 million Bitfinex status snapshots (one a minute), from which we reconstruct every funding settlement; the reconstructed rule matches the settled rates with a correlation of 0.985. BTC-perpetual funding for the perpetual-book variant is Bitfinex’s own. Table 1 summarises the sample. Volatility averaged 46.6% against a mean index of 51.9 (range 34.7–88.5); implied exceeded subsequent realised volatility on 75% of days. The daily spot-vol correlation was -0.10 overall but moved from $+0.32$ in 2023 through $+0.03$ and -0.30 to -0.59 in 2026: the market became VIX-like. Fifteen-minute index changes are autocorrelated (0.20), a trace of the index’s smoothing that matters for execution (Section 5).

Forecasts. From 15-minute log returns we compute daily realised variance (Andersen et al., 2003); fifteen minutes is coarser than the 5-minute rule of thumb (Liu et al., 2015) but matches the hedge’s frequency. Each model forecasts the average daily variance over the next $h \in \{1, 7, 30\}$ days from information at day t ’s

Moment	Live	2023	2024	2025	2026	Sim.
Realised vol, annualised (%)	46.6	43.2	53.0	44.6	44.3	51.5
Mean BVIV (vol pts)	51.9	52.4	60.2	48.3	45.2	57.1
IV – subsequent 30d RV (vol pts)	6.8					7.6
Share of days IV > RV (%)	75					78
corr(daily return, ΔIV)	−0.10	0.32	0.03	−0.30	−0.59	−0.21
corr on falling 15m bars	−0.24					−0.26
corr on rising 15m bars	0.03					0.03
Excess kurtosis, daily returns	3.4					3.7
Worst daily loss (%)	15.1					15.0

Table 1: Stylised facts: live sample (1 January 2023 to 3 October 2026) and by calendar year, against the median of the simulation’s 200 test paths (Section 6).

Model	$h = 1$ day			$h = 7$ days			$h = 30$ days		
	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t	QL ratio	MZ R^2	DM t
RW	1.71	0.16	10.47	1.33	0.15	2.56	1.42	0.09	1.70
EWMA	1.27	0.09	5.18	1.40	0.10	2.91	1.87	0.09	2.53
GARCH	1.41	0.10	8.96	1.95	0.09	5.29	3.52	0.14	3.21
HAR	1.00	0.21	–	1.00	0.19	–	1.00	0.02	–
IV-raw	1.24	0.15	6.07	1.06	0.22	0.79	0.91	0.26	−0.65
IV	1.17	0.15	2.72	1.02	0.20	0.15	0.84	0.18	−1.54
HAR-IV	0.96	0.26	−2.25	0.95	0.25	−0.74	0.92	0.16	−0.60

Table 2: Live out-of-sample forecast accuracy, 2024 to October 2026. QL ratio: a model’s QLIKE over HAR’s (best in bold). DM t : Diebold–Mariano statistic against HAR (negative favours the model).

close: a random walk, RiskMetrics EWMA (J.P. Morgan/Reuters, 1996), GARCH(1,1) with Student- t errors re-estimated monthly, the heterogeneous autoregressive (HAR) model of Corsi (2009) with a 7-day week, the index itself, raw ($IV_t^2/365$, “IV-raw”) and bias-corrected (“IV”), and HAR augmented with implied variance:

$$\ln \overline{RV}_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln \overline{RV}_{t-6:t} + \beta_m \ln \overline{RV}_{t-29:t} + \beta_{iv} \ln(IV_t^2/365) + \varepsilon_{t+h}. \quad (1)$$

HAR and IV are the special cases without the implied or the realised terms. Regressions run in logs, so forecasts $\exp(\hat{y} + s^2/2)$ are positive, and are re-estimated daily on an expanding window of targets already observed. We evaluate the 1,006 one-day forecasts from 2024 on with QLIKE, which is robust to proxy noise (Patton, 2011), the Mincer–Zarnowitz R^2 and Diebold–Mariano (DM) tests against HAR with Newey–West variance (Mincer and Zarnowitz, 1969; Diebold and Mariano, 1995).

Results. Table 2 confirms the pattern documented on Deribit data by Hoang and Baur (2020). At one day HAR-IV is best (QLIKE ratio 0.96) and significantly better than HAR (DM −2.25), while the index alone is worse (1.17) and GARCH on daily returns lags (1.41), as Bergsli et al. (2022) found for BTC. At longer horizons the differences are not significant. HAR-IV stays ahead at seven days (0.95; DM −0.74), with the bias-corrected index close behind (1.02); at 30 days the index is best (0.84; DM −1.54), followed by HAR-IV (0.92), and correcting it for the variance risk premium helps (raw IV 0.91). The calibrated simulation has the same winner at every horizon (Section 6). The 30-day HAR-IV forecast feeds the hedge sizing below.

4 Hedging framework

P&L. The book is long q BTC at price S_t and long $H_t \geq 0$ bviv-perpetual contracts with mark F_t . A position chosen at the close of 15-minute bar t earns over bar $t+1$

$$\Pi_{t+1} = q(S_{t+1} - S_t) - \mathbb{1}_{\text{perp}} q S_t f_t \Delta t + H_t(F_{t+1} - F_t) - H_t \psi_t - (H_{t+1} - H_t)(P_{t+1} - F_{t+1}) - c |H_{t+1} - H_t| F_{t+1}, \quad (2)$$

where Δt is the bar length in years, f_t the BTC perpetual’s annualised funding rate, ψ_t the bviv funding per contract, booked on the position held at each settlement, P_t the perpetual’s order-book mid, so buying above the mark is a cost, and c a proportional cost of 6 bp fee plus 10 bp slippage. On live data ψ_t is the settled funding. A perpetual with funding intensity k is priced at the expected index sampled at a random time (He et al., 2022; Ackerer et al., 2025); in the simulation we split its funding into an arbitrage-free drift transfer $\mathbb{E}_t[I_{t+1}] - I_t$ and a carry premium.

Rule	ES red.	Var red.	Max DD	Tail offset	Hedge P&L	Funding	Trading	Turnover	Notional
Unhedged	0.0	0.0	53.0	0.0	0.00	0.00	0.00	0.0	0.0
Always-on (MV)	19.0	15.3	41.0	28.0	2.46	1.42	0.34	1.3	11.1
Always-on, 1.5×	20.5	14.0	41.3	39.6	1.42	2.19	0.65	1.8	16.6
VWAP-switch	17.5	13.2	38.2	27.4	12.82	−0.29	4.56	32.3	4.6
VWAP-ratchet	20.9	15.3	41.7	31.9	3.11	2.68	1.21	2.3	12.5

Table 3: Hedging a 1-BTC spot book with the Bitfinex bviv perpetual, 3 April 2024 to 3 October 2026 (% , except turnover). ES/Var red.: reduction in daily ES_{97.5} and variance against unhedged. Max DD: maximum drawdown. Tail offset: hedge P&L net of costs as a share of BTC losses on the worst 5% of days. Hedge P&L (index P&L minus funding), Funding and Trading (fees, slippage and premium): % of BTC notional a year. Turnover: perpetual notional traded per year over BTC notional. Notional: average hedge notional in % of BTC.

Hedge ratios and signal. The minimum-variance hedge is $H_t^* = -qS_t\beta_t$, with $\beta_t = \text{Cov}_t(r, \Delta F) / \text{Var}_t(\Delta F)$ estimated on 15-minute changes by an EWMA (half-life 14 days) (Johnson, 1960; Ederington, 1979); we impose $H \geq 0$. A downside ratio $\beta_t^- = \mathbb{E}_t[r \Delta F \mid r < 0] / \mathbb{E}_t[\Delta F^2 \mid r < 0]$ on falling bars, shrunk towards β_t , gives the crash hedge H_t^- . The breakdown statistic compares the price with its rolling 24-hour VWAP,

$$z_t = \ln(S_t / \text{VWAP}_t^{24h}) / (\hat{\sigma}_t \sqrt{1/3}), \quad (3)$$

because for Brownian prices the gap between the last price and its average over a window w has standard deviation $\sigma\sqrt{w/3}$. Here $\hat{\sigma}_t$ is a causal EWMA (half-life one day) of 15-minute squared returns, de-seasonalised by a trailing hour-of-week profile (Andersen and Bollerslev, 1997; Hansen et al., 2024).

Rules. All rules are long-only, capped at 50% of BTC notional, and trade only when the target leaves a 25% band around the position (Whalley and Wilmott, 1997). *Always-on* holds λH_t^* . The *VWAP-switch* holds H_t^- while a gate is on, switching on at $z_t < z_{in}$ and off at $z_t > z_{out}$. The *VWAP-ratchet* holds a core ϕH_t^* plus an overlay o_t that jumps to 1 on a breakdown and decays with half-life τ :

$$o_t = \begin{cases} 1 & z_t < z_{in} \\ 2^{-1/(96\tau)} o_{t-1} & \text{otherwise,} \end{cases} \quad H_t = \phi H_t^* + o_t \max(H_t^- m_t - \phi H_t^*, 0), \quad (4)$$

where $m_t \in [0.5, 1.5]$ optionally scales the overlay by forecast “cheapness”, the log ratio of the 30-day HAR-IV forecast to implied variance.

Protocol. Rule parameters are *frozen*: they were selected on 48 simulated training paths (Section 6), out of 72 ratchets and six switches, by median ES reduction net of costs. The selected ratchet keeps the full core ($\phi = 1$), triggers at $z < -1.5$, decays with a half-life of 4 days and uses forecast sizing; the selected switch enters at $z < -1$ and exits at $z > 0.5$. The live sample is therefore pure out-of-sample evidence, which avoids the snooping trap of choosing among many rules on the evaluation data (White, 2000; Hansen, 2005; Bailey et al., 2014). Signals warm up on 2023 data, and hedging is evaluated over the 914 days on which the perpetual traded, 3 April 2024 to 3 October 2026. On a single path we report 95% intervals from a paired stationary block bootstrap over days (mean block ten days, 2,000 draws). The timing placebo shifts the ratchet’s trigger series circularly by every multiple of a week, which keeps the triggers’ number, clustering and hour-of-week profile but breaks their link to the market.

5 Results on live data

The hedge works, in crashes. The minimum-variance hedge reduced ES by 19.0% (block-bootstrap CI 6.9–33.3%) and variance by 15.3% (4.8–27.8%), and cut the maximum drawdown from 53.0% to 41.0% with an average notional of 11.1% of the BTC position (Table 3); it was held on 67% of bars and switched off whenever the estimated spot-vol covariance turned positive. It is a crash hedge in the literal sense (Figure 2). Net of funding, its P&L summed to +6.2% of BTC notional: the ten best days contributed 53% and all other days −46%. On the ten worst BTC days, which cost the unhedged book 74%, the hedge earned back 38%. The best day, 5 February 2026, returned 21% of BTC notional, when bviv rose from 57 to 96 within hours as BTC fell by 14%. Figure 3 shows that day at 15-minute resolution. BTC broke through its VWAP band at 14:00 UTC, after bviv had jumped from 61 to 74 in two hours; it reached 96 by 21:45. All rules entered the day with 560–610 contracts. As the index’s 15-minute moves exploded, the EWMA hedge ratio fell and the rules sold into the spike, which kept the hedge’s dollar risk roughly constant; the ratchet’s overlay kept

Always-on hedge: cumulative P&L (% of BTC notional)

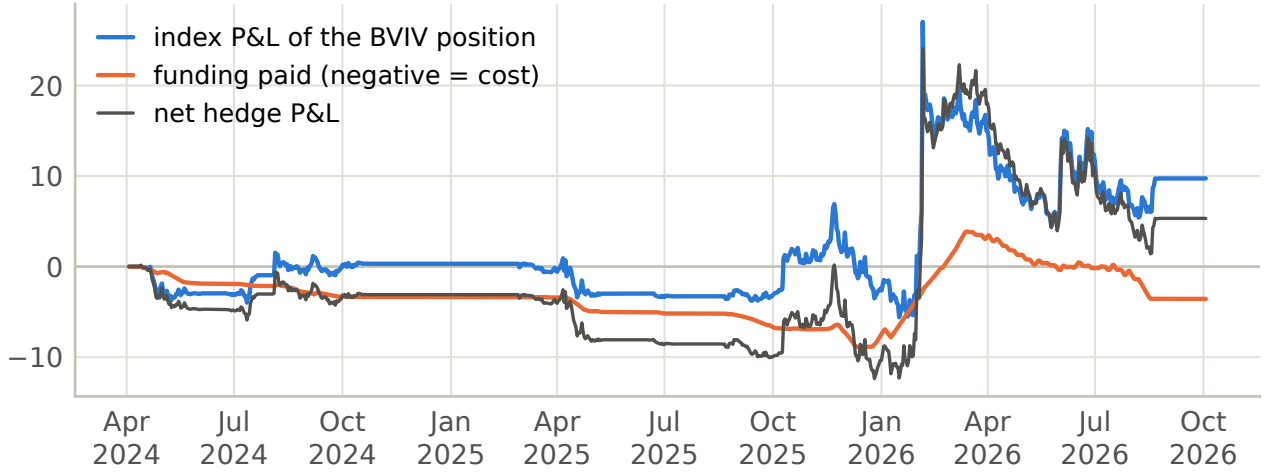
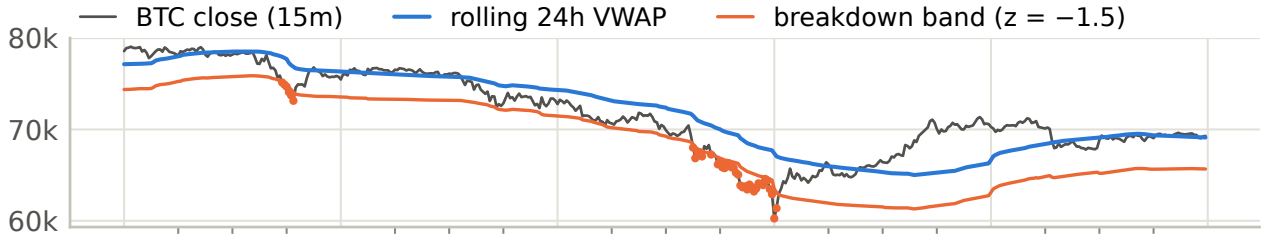


Figure 2: Always-on minimum-variance hedge on live data: cumulative index P&L of the BVIV position, funding paid and net hedge P&L, in % of BTC notional.

BTC, its VWAP and the breakdown band (USD)



BVIV (vol points)



Hedge size for 1 BTC (contracts, \$1 per vol point)

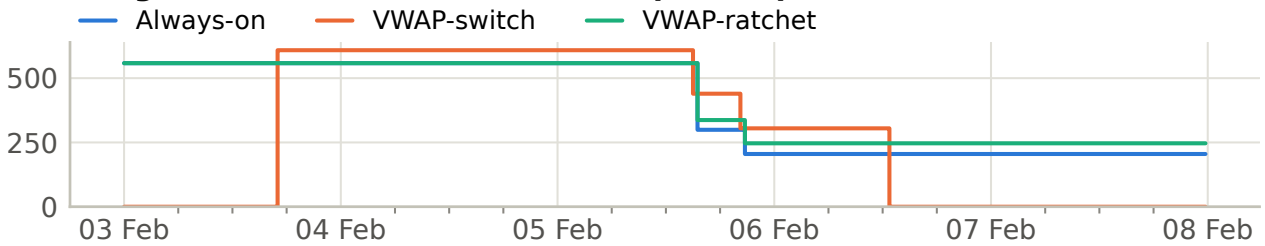


Figure 3: The best hedge day, 5 February 2026, at 15-minute resolution (UTC). Top: BTC, its rolling 24-hour VWAP and the ratchet's breakdown band (dots: $z_t < z_{in}$). Middle: the Volmex index and the Bitfinex perpetual's mid. Bottom: hedge sizes of the three rules for a 1-BTC book.

its position 13–20% above the always-on hedge's. Funding cost the hedge 1.4% of BTC notional a year on average: it paid 4.5% a year in 2024 and 4.3% in 2025 and received 5.4% in 2026.

Timing adds size, not information. The switch traded 32.3 times the BTC notional a year and protected less than the static hedge (paired ES difference -1.44 pp, CI $[-8.09, +2.54]$). The ratchet beat the always-on hedge by $+1.94$ pp of ES reduction (CI $[-0.48, +4.08]$) but not the $1.5\times$ static hedge ($+0.39$ pp, CI $[-1.96, +2.81]$). Against its 194 placebos it ranks at the median: it beats 50% of them, whose ES reductions span 20.4–21.3% (Figure 4, left). Across the 72-configuration grid the timing value, the ES reduction minus the placebo mean, averages $+7.82$ pp without a core hedge (rules beat a mean 98% of their placebos), $+1.72$ pp with half a core and $+0.78$ pp with the full core (90%; right panel). VWAP breakdowns carry information

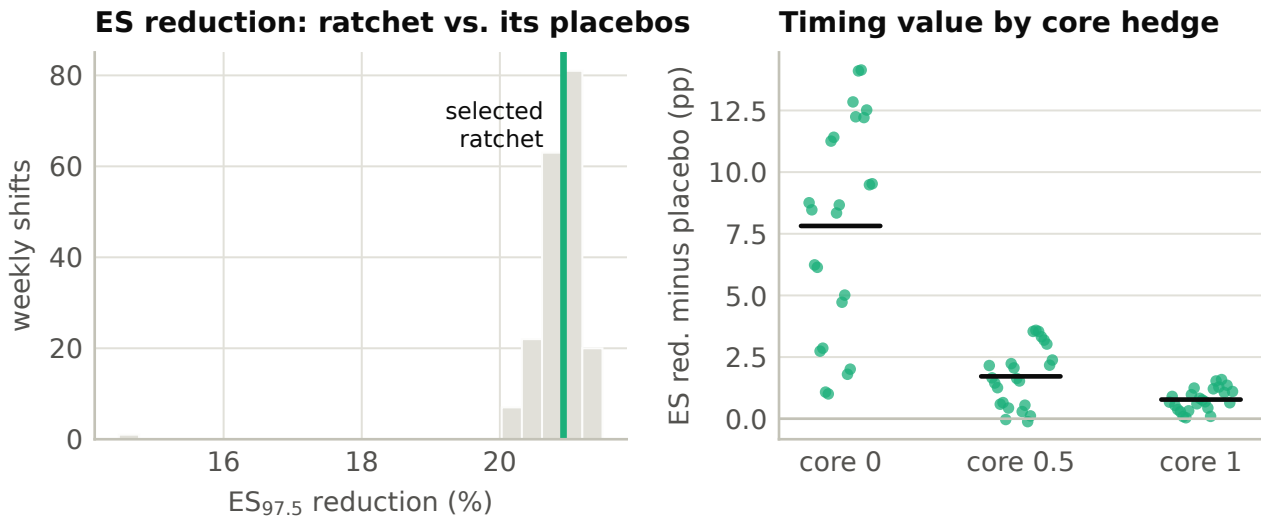


Figure 4: Timing placebos on live data. Left: ES reduction of the selected ratchet (line) against all weekly circular shifts of its triggers. Right: timing value of each of the 72 ratchet configurations by core size; bars mark the means.

Variant	Always-on	1.5×	Switch	Ratchet
Execution 1 bar later	17.6 (+2.0)	19.7 (+0.8)	16.4 (+6.7)	20.0 (+1.4)
Execution 1 hour later	15.8 (+0.5)	17.0 (−1.8)	12.1 (+2.2)	17.9 (−1.6)
Execution 1 day later	11.8 (−1.4)	10.4 (−5.4)	9.0 (+0.9)	12.6 (−2.6)
No trading costs	19.0 (+2.5)	20.6 (+1.4)	17.7 (+12.8)	21.0 (+3.1)
Trading costs ×2	18.9 (+2.5)	20.5 (+1.4)	17.4 (+12.8)	20.9 (+3.1)
Funding ignored	19.0 (+3.9)	20.5 (+3.6)	17.6 (+12.5)	21.0 (+5.8)
Fills at mark (no premium)	19.0 (+2.5)	20.4 (+1.4)	20.1 (+12.8)	21.1 (+3.1)
Perpetual BTC book	19.1 (+2.5)	20.6 (+1.4)	17.6 (+12.8)	21.0 (+3.1)

Table 4: Live-data sensitivities, frozen rules: ES reduction in % (hedge P&L in % of BTC notional a year).

about the next crash, but a core hedge already captures most of it.

Execution and capacity. Table 4 varies one assumption at a time. Trading costs and funding barely move the ES reductions, since they are small next to crash-day gains; only the switch, which trades often, gains from fills at the mark (20.1% instead of 17.5%). Execution delay matters most: executing one bar, one hour or one day after the signal lowers the always-on ES reduction to 17.6%, 15.8% and 11.8%. The switch’s high hedge P&L (+12.82% a year) shrinks to +2.20% with a one-hour delay. It comes from the index’s smoothing: a position opened right after a BTC drop collects the index’s catch-up. That gain requires trading within minutes against quotes that follow a lagging index, and we do not count on it. Capacity is the binding constraint. A 1-BTC book needs a median 248 contracts, against a median open interest of 84; the hedge would have exceeded the whole open interest on 61% of bars. The book is deeper than the trading: on 4 October 2026 it offered 384 contracts 0.17% above the mid (spread 0.34%) and 1,667 within 2.5%, the hedge of about 7 bitcoins.

6 The controlled experiment

A single path cannot separate rules reliably, so we repeat the study in a simulated market calibrated to the stylised facts (Table 1, last column). Log-variance is the sum of a slow and a fast Ornstein–Uhlenbeck factor; the fast factor tracks a three-state hidden regime (calm, stress, euphoria) and receives EGARCH-type shocks (Nelson, 1991) whose signed loading on the return shock is $-0.30/-0.60/+0.25$ by regime, so volatility rises in sell-offs and, in euphoria, in rallies. Jumps cluster in stress; volatility and volume follow intraday and weekend seasonality. The index is the square root of the model’s exact expectation of 30-day average variance (a Feynman–Kac product over regime paths), scaled by a variance premium and perturbed by persistent sentiment. The perpetual’s funding is the exact one-bar drift transfer plus a carry of 6 vol points a year, so it offers no free lunch from mean reversion; unit tests check both expectations against brute-force simulation. Each path has 900 days; rules are selected on 48 training paths and evaluated on 200 test paths.

Rule	ES red.	Var red.	Max DD	Tail offset	Carry	Trading	Turnover	Time on
Unhedged	0.0	0.0	49.5	0.0	0.00	0.00	0.0	0
Always-on (MV)	4.2	4.1	48.7	5.9	0.74	0.14	0.9	93
Always-on, 1.5×	4.9	3.1	49.5	8.8	1.11	0.20	1.3	93
VWAP-switch	2.8	1.9	50.9	4.9	0.38	3.94	24.6	35
VWAP-ratchet	4.9	3.9	48.9	7.8	0.96	0.23	1.4	98
Ratchet, placebo timing	4.9	3.8	48.8	7.7	0.96	0.23	1.5	98
Oracle (infeasible)	2.3	2.3	48.1	3.5	0.11	0.27	1.7	9

Table 5: Simulation: hedging a long-BTC book on 200 test paths (medians; % unless stated). Columns as in Table 3; Carry: funding premium. Placebo timing: the selected ratchet with circularly shifted triggers (mean of 5 shifts).

Scenario	ES red. (%)		Ratchet – Always		Switch > 0	Cost (% p.a.)		
	Always	Ratchet	mean [95% CI]			ES red.	Always	Ratchet
Baseline	4.3	5.1	+0.66 [+0.39, +0.94]		71%	3.6	0.87	1.22
Trading costs ×2	4.3	5.1	+0.63 [+0.37, +0.91]		71%	3.4	1.01	1.46
Trading costs ×0.5	4.3	5.2	+0.67 [+0.41, +0.95]		71%	3.7	0.80	1.10
Carry 0 vol pts	4.3	5.2	+0.67 [+0.40, +0.95]		71%	3.7	0.13	0.24
Carry 12 vol pts	4.3	5.1	+0.65 [+0.38, +0.93]		71%	3.6	1.61	2.20
Carry 24 vol pts (live avg.)	4.2	5.0	+0.63 [+0.36, +0.91]		71%	3.6	3.10	4.15
VIX-like spot-vol	9.7	10.2	+0.74 [+0.47, +1.03]		69%	6.3	1.33	1.64
Inverse leverage	0.5	0.7	+0.18 [−0.02, +0.39]		58%	0.5	0.25	0.57
Crashes cluster in stress	5.3	5.1	+0.34 [−0.00, +0.68]		61%	3.1	0.85	1.18
Session-anchored VWAP	4.3	5.4	+0.73 [+0.44, +1.02]		73%	3.4	0.87	1.25
Perpetual BTC book	4.3	5.1	+0.64 [+0.38, +0.92]		71%	3.6	0.87	1.22

Table 6: Simulation robustness on 96 paths with common random numbers: medians, and the paired mean ES difference between the ratchet and the always-on hedge with a 95% bootstrap CI; > 0: share of paths on which the ratchet wins.

Findings. The forecast ranking is the same as on live data: HAR-IV is best at one day (median QLIKE ratio 0.95) and seven days (0.89), the bias-corrected index at 30 days (0.85). The hedging results have the live pattern at a smaller scale (Table 5), because the simulated spot-vol correlation (−0.21) is weaker than in 2025–26. The always-on hedge reduces ES by 4.2%; the switch trails it by 1.67 pp. The ratchet beats it by +0.53 pp (95% CI [+0.33, +0.72]) but its placebo by only +0.05 pp, and a 1.5× static hedge reaches the same ES reduction (4.9%). Across the grid, timing adds +0.71 pp without a core and +0.01 pp with the full core, the same gradient as on live data. The ratchet’s edge over the always-on hedge (Table 6) survives doubled or halved trading costs and a carry of 0 to 24 vol points a year (+0.63 to +0.67 pp; +0.63 pp at the live average of 24). It shrinks to +0.18 pp under inverse leverage and to +0.34 pp when crashes cluster in stress.

7 Limitations and conclusion

The live sample is one path of 914 days, whose tail is dominated by a few crashes, and the hedge’s gains depend on them; the block bootstrap quantifies but cannot remove this. Results assume fills at the 15-minute close at the observed mid plus costs, which the book supports only for a few bitcoins, and the delay results show how much depends on prompt execution. Oracle risk on builder-deployed venues (Hyperliquid, 2026) and the 15-minute boundary’s algorithmic order flow (Kim and Hansen, 2026) argue for working orders inside the bar, for example by VWAP slicing (Berkowitz et al., 1988; Konishi, 2002; Białkowski et al., 2008). The simulation’s spot-vol dynamics are calibrated, not estimated.

Four lessons follow. (1) Forecast BTC volatility with HAR augmented with implied variance up to a week, and with the bias-corrected implied index at a month. (2) A minimum-variance BVIV hedge cut tail risk substantially in 2024–26, because the market turned VIX-like and crashed with spiking implied volatility; it should be held, not switched. (3) Its price is funding, which on Bitfinex is capped, volatile and usually paid by longs, and its limit is capacity: the perpetual traded a median \$71 a day, and its book held the hedge of a few bitcoins. (4) A 15-minute VWAP breakdown is real but secondary information: it beats random timing without a core hedge, but with one a ratchet’s gain is size, not timing. Any intraday timing rule should beat a placebo with the same trigger statistics before it earns its complexity.

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